

Strike while the iron is hot – optimal monetary policy under state-dependent pricing

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Central question of monetary economics: the nature of nominal rigidities

- **“Old” time-dependent (TD) models** (Calvo, Taylor):
tractable, give the New Keynesian Phillips curve

$$\pi_t = \kappa \cdot \widehat{mc}_t + \beta \mathbb{E}_t[\pi_{t+1}] \quad (\text{NKPC})$$

- **“New” state-dependent (SD) models** (menu cost: Golosov–Lucas, Nakamura–Steinsson):
match the rich micro evidence on price changes

Main results:

- After *large* cost-push shocks, optimal policy leans against inflation *disproportionately* hard
- Because repricing frequency rises endogenously $\Rightarrow$ prices more flexible $\Rightarrow$ lower sacrifice ratio
- TFP shocks: optimal policy still fully stabilizes inflation (“divine coincidence”)
- Optimal long-run inflation: slightly positive (0.07% p.a.), not exactly zero

Model I

Standard textbook NK model, but with state-dependent pricing.

Household problem:

$$\max_{\{C_t, N_t, B_t\}} \sum_{t=0}^{\infty} \beta^t [\log C_t - \chi N_t]$$

s.t.

$$P_t C_t + Q_t B_t + T_t = B_{t-1} + W_t N_t + D_t$$

Final goods producer is subject to Dixit-Stigitz aggregator:

$$Y_t = \left\{ \int_0^1 [A_t(j) Y_t(j)]^{\frac{\epsilon-1}{\epsilon}} dj \right\}^{\frac{\epsilon}{\epsilon-1}}$$

where $A_t(j)$ is the idiosyncratic quality of good j

Model II

Quality is leptokurtic (fat tailed)

$$\log A_t(i) = \log A_{t-1}(i) + \varepsilon_t(i), \quad \varepsilon_t(i) \sim \begin{cases} N(0, \sigma_1^2) & \text{with prob } \hat{\omega} \\ N(0, \sigma_2^2) & \text{otherwise} \end{cases}$$

Intermediate good producers have production functions:

$$Y_t(j) = A_t \frac{N_t(j) M_t(j)^{1-\alpha}}{A_t(j)} \quad (1)$$

where $M_t(j)$ is final good used as intermediate input. Strategic complementarity:

$$MC_t(j) = \frac{A_t(j)}{A_t \alpha^\alpha (1-\alpha)^{1-\alpha}} W_t^\alpha P_t^{1-\alpha} \quad (2)$$

Model III

Period t profits are:

$$D_t(j) = [P_t(j) - (1 - \tau_t)MC_t(j)]Y_t(j)$$

subsidy:

$$\tau_t - \tau = \rho_\tau(\tau_{t-1} - \tau) + \tau\epsilon_{\tau,t} \quad (\text{cost-push shock})$$

$$\log A_t = \rho_A \log A_{t-1} + \epsilon_{A,t} \quad (\text{TFP shock})$$

Firm pays "menu" cost η in units of labor to change its price

Menu cost model solution I

Solution of canonic menu cost model.

Define price gap:

$$x_t(j) = \log P_t(j) - \log P_t^*(j) = p_t(j) - p_t^*(j)$$

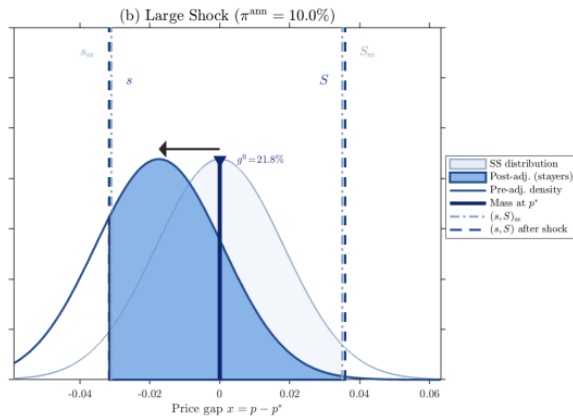
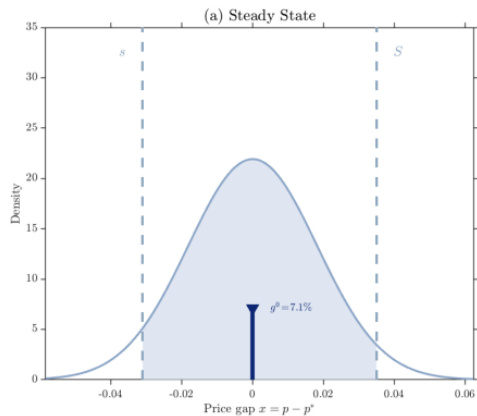
$$x_t = x_{t-1} - \epsilon_t - \pi_t^*$$

$$\pi_t^* = p_t^* - p_{t-1}^* + \pi_t$$

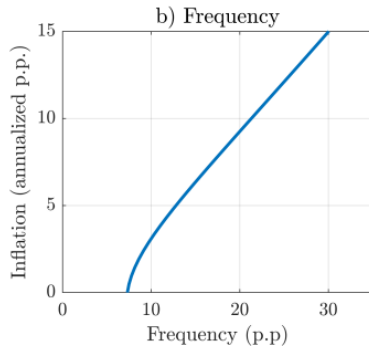
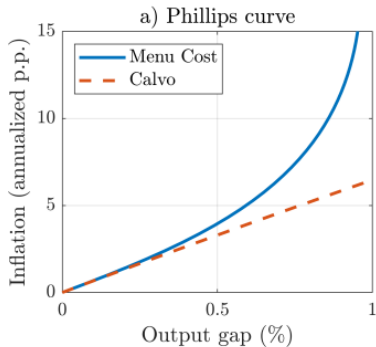
Let $V_t(x_t)$ firm's end-of-period value function expressed in terms of the price gap.

Solution? Change price to p_t^* ($x_t = 0$) if $x_t \notin [s_t, S_t]$ (sS bound).

Menu cost model solution II



Optimal policy I

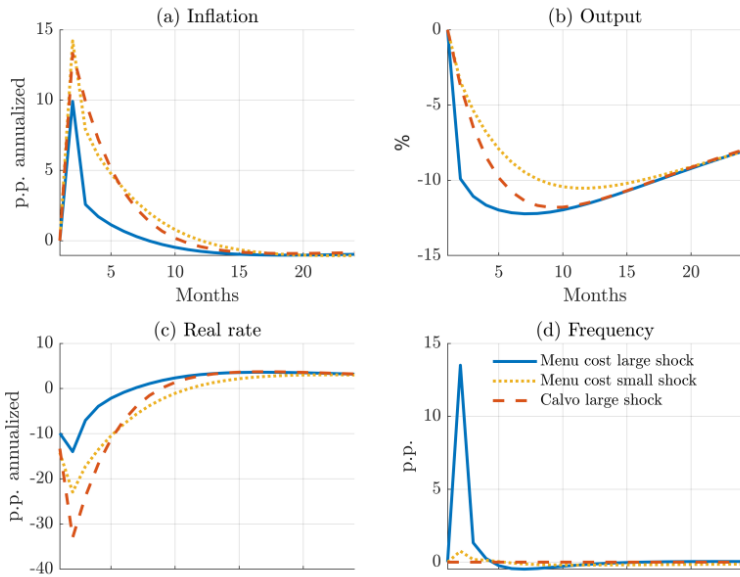


Optimal policy II

Optimal policy with Calvo:

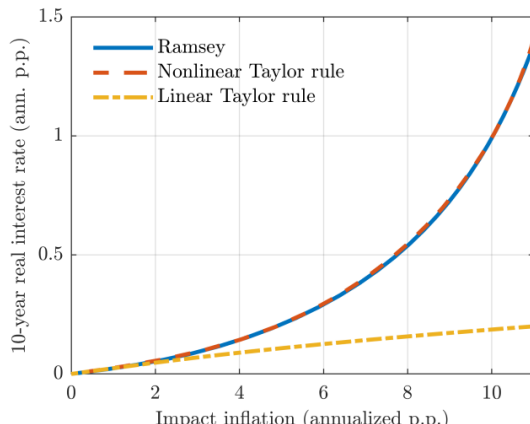
$$\begin{aligned} & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\pi_t^2 + \frac{\kappa}{\varepsilon} x_t^2 \right] \\ & \pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa x_t, \quad \text{where } \kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta} \\ & \pi_t + \frac{1}{\varepsilon} \Delta x_t = 0 \end{aligned} \tag{3}$$

Optimal policy III



Optimal policy IV

$$\log(R_t/R) = \rho_i \log(R_{t-1}/R) + (1 - \rho_i) \{ \underbrace{\phi_\pi(\pi_t)\pi_t}_{\phi_{\pi,1} + \phi_{\pi,2}\pi_t^2} + \phi_y(y_t - y_t^*) \}$$



Optimal policy V

